

INFO703 BIG DATA AND ANALYTICS

AROMATAWAI TUATAHI | ASSIGNMENT ONE

Due: Week	7 (Friday 23:59pm)
Weightings:	30% of the final grade
Task:	Data Analysis and Classification

Learning outcome:

- 1 - Examine application architectures for big data and analytics
- 2 - Examine and apply data modeling approaches for data mining techniques
- 4 - Demonstrate an understanding of commonly used industry tools

Guidelines

Assignment 1 aims to enhance your proficiency in data cleaning and modeling. You'll select a dataset either from the provided list or through your own search. Utilising a data mining tool such as Weka or RapidMiner, you'll explore different classification techniques. Following this experimentation, you'll compile a report detailing your findings and results. The report must have 1500 words.

1. Choosing a Dataset

Several datasets have been provided for you and uploaded in Moodle (Lab 1). The datasets are briefly described. Each student should choose one of these datasets for the assignments. **By the end of week 3**, each person needs to confirm to the tutor the selection of the dataset.

You are recommended to consult with your tutor to choose an appropriate dataset. You can continue your work on the dataset through next assignments, should you wish to.

Please note, the data set must have at least 20,000 records and a minimum of 12 attributes after data preparation phase.

Note: Each dataset can be chosen by one person only in first come first serve basis. Any dataset outside the list can be also considered, but must be approved by the tutor. This **MUST BE DONE BY END OF WEEK 3**.

2. Dataset Description

After choosing the dataset, you need to ensure that the dataset is ready for your classification task (data preparation). Double check if there is any missing value or outlier, fix it before starting the analysis.

You may need to modify the dataset to make it suitable for classification. For example, you may need to consider one of the attributes as the class attribute and, if needed, discretize the value of this attribute.

In your report, explain clearly what preparation steps have you taken? (Note: After this stage, there must be at least 20,000 records remained in the data set).

Then, measure some statistical factors about the dataset using your choice of data mining tool: Min, Max, Standard Deviation, Mean. For non-numerical attributes, digitize it and provide relevant values if applicable. Then, provide appropriate charts to visualize your data.

You need to provide at least 8 charts (of different types). In your report, explain each graph and its significance to your analysis.

3. Classification

After preparing your dataset, you will experiment with **four** different classification techniques (of your choice) supported by Weka/RapidMiner or another data mining tool and see which one gives you better results for your dataset. You need to experiment with classifiers and measure different metrics as depicted in the following table.

Each classification technique must be applied to the dataset with two different sizes of training set (60% and 80%).

Analyse your results and discuss which techniques better suit your dataset?

Note: Your classifiers may be different from the ones shown in the table.

	Training Set Size	TP Rate	FP Rate	Accuracy
J48 Tree	60%			
	80%			
Simple Cart	60%			
	80%			
Naïve Bayes	60%			
	80%			
AD Tree	60%			
	80%			

DELIVERY

1. A Zip file containing the following items must be submitted online by Week 7, Friday, 5pm (no late submission please).
2. A document file (.doc, .docx, or .pdf) for the report which includes the following sections:
 - a. Description of the dataset and why did you choose it?
 - b. The steps you have taken for data preparation. Explain in detail the reasons for any changes to the attributes for analysis.
 - c. Statistical parameters for the dataset and Charts
 - d. The table for the results of classification
3. The modified dataset file

Submissions missing any of the deliverables mentioned above will be marked as 0.

If you have any question or confusion about the instructions, please seek help from your tutor as soon as possible.

MARKING SCHEDULE

Assessment Against Criteria			
Criteria	Marks	Score	
Report			
Choosing a Dataset			/8
	Chosen by Week 3	8	
Data Description			/20
	Explanation of the Dataset and its relevance and possible uses	10	
	Explaining attributes	10	
Data Preparation and Visualization			/45
	Handling missing values	10	
	Handling outliers	5	
	Choice of an appropriate class attribute	5	
	Discretising the class attribute (if needed)	5	
	Measuring statistical metrics (Any five)	10	
	Drawing different charts and explaining each	10	
Classification			/27
	Measuring required metric for the four classifiers	15	
	Justify the best classifier for your dataset	12	
TOTAL		100	